

W 13 L 01

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Time Complexity.

NP-completeness.

$$\text{SAT} = \{ \langle \phi \rangle \mid \phi \text{ is a satisfiable formula} \}$$

ϕ Boolean formula. in n variables

$$\phi: \{0,1\}^n \rightarrow \{0,1\}$$

$$\phi(x,y) = \bar{x} \vee y$$

we say a Boolean formula is satisfiable

iff there exists a truth assignment to its variables st. it makes it true //1.

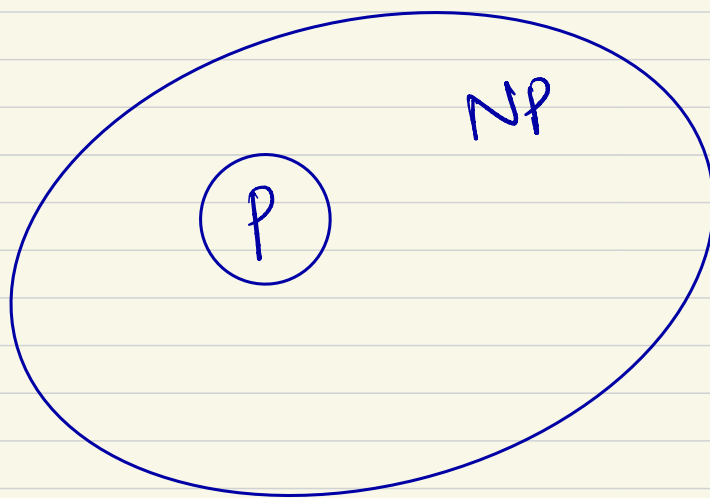
$$\text{if } x=1, y=0$$

whether some formula $\varphi_1 \in \text{SAT}$

Question: is SAT decidable? yes

claim: $\text{SAT} \in \text{NP}$

Theorem: $\text{SAT} \in \text{P}$ iff $\text{P} = \text{NP}$.

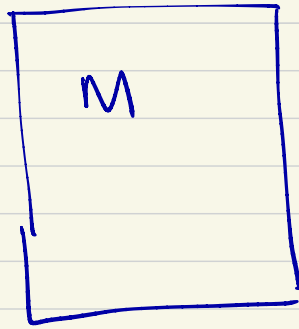


Polynomial Time Reducibility

A function $f: \Sigma^* \rightarrow \Sigma^*$ is a

polynomial time computable function if
some polynomial time TM M exists
that halts with just $f(w)$ on its
tape, when started on input w .

$$f: \Sigma^* \rightarrow \Sigma^*$$



$$g: \Sigma^* \rightarrow \Sigma^*$$

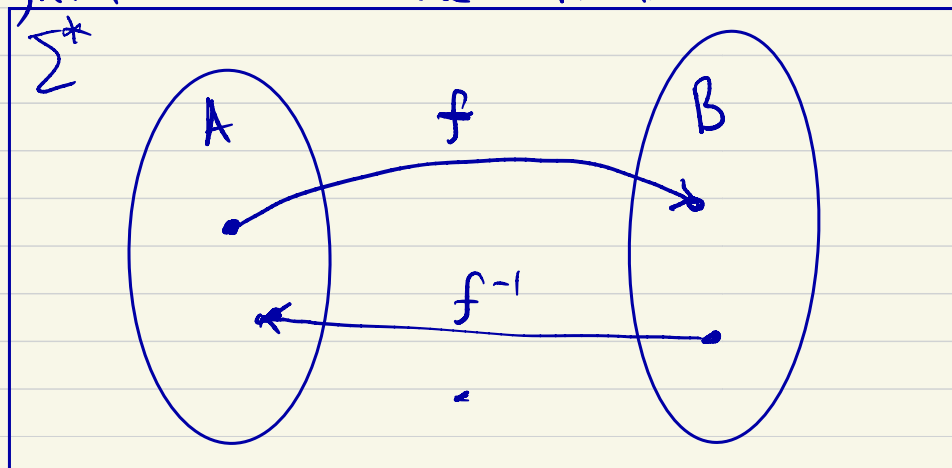
$$g(w)$$

output all possible substrings of w .

A language A is polynomial time mapping reducible // polynomial time reducible, to language B , written as $A \leq_p B$ if a polynomial time computable function $f: \Sigma^* \rightarrow \Sigma^*$ exists, where for every w :

$$w \in A \iff f(w) \in B.$$

The function f is called polynomial time reduction.



Theorem: if $A \leq_p B$ and $B \in P$, then $A \in P$.

Definition

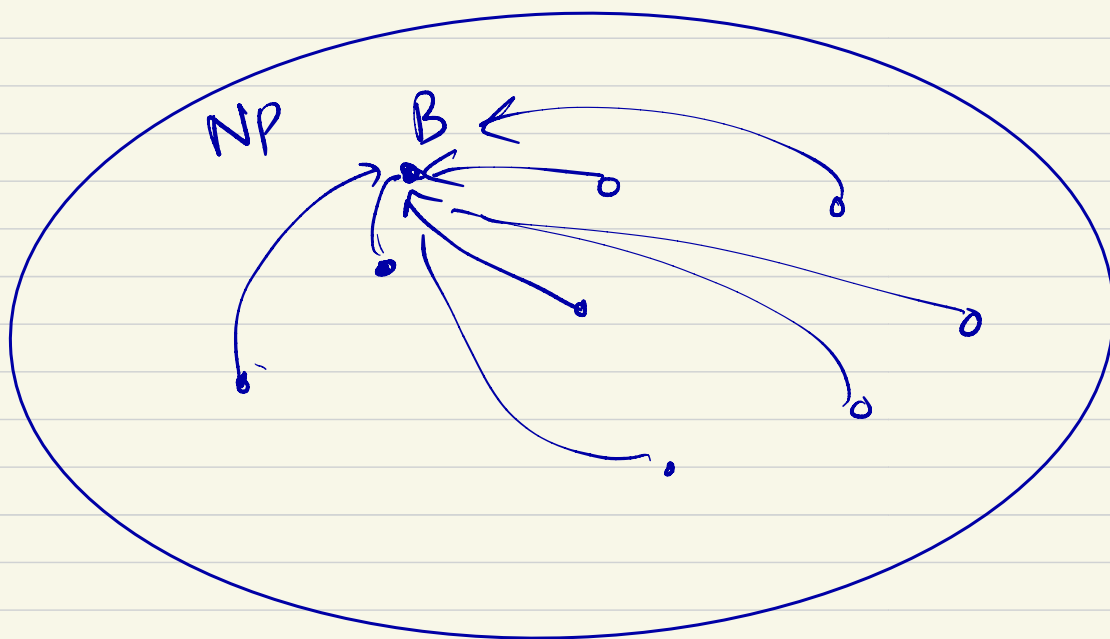
A language B is NP-complete if it satisfies two conditions.

(1) B is in NP // $B \in NP$

(2) every A in NP is polynomial time reducible to B .

// $\forall A \in NP, A \leq_p B$.

B is NP-complete



Theorem: If B is NP-complete and $B \in P$ then $P = NP$.

Proof:

Theorem

if B is NP-complete and $B \leq_p C$ for some C in NP, then C is NP-complete

$A \leq_p B$ and $B \leq_p C$ then $A \leq_p C$

```
{ String f (string w) {  
    :  
    return some-string;  
}
```

```
{ String g (string y) {  
    :  
    return some-string;  
}
```

```
{ String h(string w) {  
    w' = f(w);  
    w'' = g(w');  
    return w'';  
}
```

